

**ILOILO STATE UNIVERSITY OF FISHERIES SCIENCE
AND TECHNOLOGY**

**COLLEGE OF INFORMATICS AND COMPUTING
INNOVATIONS**

Software Design Description
for
School Event Management System

Signature

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Change History

Version	Description of Changes	Author
1.0	Initial Software Design Description	Project Team
1.1	Revision of system design	Project Team
1.2	Final revision	Project Team

Preface

This Software Design Description (SDD) presents the proposed technical and structural design of the School Event Management System for the Iloilo State University of Fisheries Science and Technology. The document follows the organization of the provided SDD template, which includes introduction, references, decomposition description, dependency description, interface description, detailed design, appendices, index, and annexes. □filecite□turn1file0□L57-L177□

The SDD translates the system requirements into a design that can guide development. It identifies the major software modules, processes, data entities, interfaces, dependencies, and detailed design considerations for the proposed event-management application.

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1. Introduction

1.1. Purpose

The purpose of this Software Design Description is to define the technical design of the School Event Management System. It provides a blueprint for implementing the system's major modules, processes, data structures, interfaces, and dependencies.

1.2. Scope

The design covers a web-based system for managing school events from event proposal and approval through publication, participant registration, attendance recording, announcements, and reporting. The design includes administrator, event organizer, student/participant, and faculty/staff access.

The initial design does not include financial accounting, academic-record management, or other university processes outside school-event management.

1.3. Definitions and Acronyms

SDD – Software Design Description

SEMS – School Event Management System

Admin – Authorized system administrator

Organizer – Authorized person who creates and manages events

Participant – Registered user who joins an event

UI – User Interface

DB – Database

CRUD – Create, Read, Update, Delete

2. References

The primary structural reference for this document is the uploaded SDD template from the College of Information and Communication Technology. The template specifies the sequence of sections from Introduction through Annexes, including module, process, data, dependency, interface, detailed-design, and diagram sections. □filecite□turn1file0□L57-L177□

Project requirements are based on the previously defined School Event Management System concept. Specific implementation technologies are not fixed by the template and should be confirmed by the project team before development.

3. Decomposition Description

3.1. Module Decomposition

The system is divided into functional modules so that each major responsibility can be developed and maintained independently while sharing controlled data through the application database.

Module	Primary Responsibility	Main Users
User and Access Management	Authentication, authorization, user profiles, account status	Admin, all users
Event Management	Event creation, editing, approval, publication, scheduling	Admin, Organizer
Registration and Attendance	Participant registration, capacity checking, participant lists, attendance	Organizer, Participant, Admin
Notification and Reporting	Announcements, event updates, summaries, participant and attendance reports	Admin, Organizer, Participant

3.1.1. User and Access Management Module

Provides login, logout, role-based authorization, account creation, account updates, and profile management. It is responsible for ensuring that protected functions are accessible only to authorized users.

3.1.2. Event Management Module

Provides event creation, editing, submission, approval, scheduling, publication, status management, and event-detail viewing.

3.1.3. Registration and AttendanceModule

Handles participant registration, registration validation, event capacity, participant lists, attendance recording, and attendance summaries.

3.1.4. Notification and Reporting Module

Manages event announcements and relevant notifications and produces basic event, registration, and attendance reports.

3.2. Concurrent Process Decomposition

3.2.1. Authentication Process

The authentication process accepts user credentials, validates them against stored account information, establishes an authenticated session when valid, and assigns access according to the user's role. Invalid credentials result in an error message without granting protected access.

3.2.2. Event Registration and Attendance Process

Registration checks event availability, participant eligibility, and duplicate registration before creating a registration record. Attendance processing retrieves registered participants and records attendance for the selected event. These activities can occur while other users browse or manage unrelated events.

3.3. Data Decomposition

3.3.1. User Data Entity

Stores user identification, account credentials or password hash, role, contact information where required, and account status.

3.3.2. Event Data Entity

Stores event title, description, category, date, time, venue, organizer, capacity, registration period, status, and timestamps.

3.3.3. Registration Data Entity

Associates a participant with an event and stores registration status and registration timestamp.

3.3.4. Attendance Data Entity

Associates a participant and event with attendance status and attendance timestamp.

4. Dependency Description

4.1. *Inter-module Dependencies*

User and Access Management is a foundational dependency because Event Management, Registration and Attendance, and Reporting functions require authenticated and authorized users. Event Management supplies event records required by Registration and Attendance. Registration records are required for participant lists and attendance reporting.

4.2. *Inter-process Dependencies*

The registration process depends on successful authentication and on an available approved event. Attendance processing depends on an existing event and, where applicable, an existing participant registration. Reporting depends on the availability and integrity of stored event, registration, and attendance records.

4.3. *Data Dependencies*

User records are referenced by organizer and participant relationships. Event records are referenced by registrations and attendance records. Registration records provide the participant-event relationship used when generating participant lists and attendance summaries.

5. Interface Description

5.1. Module Interface

5.1.1. User and Access Management Interface

Inputs: username/email and password, account details, role and account-status changes. Outputs: authentication result, user profile, authorization status, and account-management confirmations.

5.1.2. Event Management Interface

Inputs: event details, proposed schedule, venue, capacity, organizer information, approval decision, and event status changes. Outputs: event record, approval status, published event information, and validation messages.

5.1.3. Registration and Attendance Interface

Inputs: event selection, participant information, registration action, attendance action. Outputs: registration confirmation/status, participant list, attendance status, and attendance summary.

5.2. Process Interface Process 1 Description

5.2.1. Authentication Interface

The authentication interface connects the login UI with the user/account data service. It returns either an authenticated session and role or an authentication failure.

5.2.2. Event Registration Interface

The registration interface connects event details and participant actions with the registration data service. It checks event status, registration period, capacity, and duplicate registration before saving the transaction.

6. Detailed Design

6.1. Module Detailed Design

6.1.1. User and Access Management Detail

- Login: validate credentials and establish session.
- Role control: determine permitted functions from the user's role.
- User management: allow administrators to create, update, deactivate, and review accounts.
- Logout: terminate the authenticated session.

6.1.2. Event Management Detail

- Create event: organizer enters required event details and saves a proposal.
- Review event: administrator checks the proposal and approves or rejects it.
- Publish event: approved events become visible to intended participants.
- Update event: authorized users modify permitted details and system records the new status or timestamp.
- Cancel event: authorized users mark an event as cancelled and display the updated status.

6.1.3. Registration and Attendance Detail

- Registration validates the participant and event.
- Capacity validation prevents registration beyond the configured limit.
- Duplicate validation prevents multiple active registrations for one participant and event.
- Attendance records the participant's attendance status for the selected event.

6.1.4. Notification and Reporting Detail

- Announcements are associated with a specific event and made available to authorized viewers.
- Reports query event, registration, and attendance data.
- Event changes may trigger an in-system notification to affected participants.

6.2. Data Detailed Design

6.2.1. User Data Detail

UserID, FullName, Email/Username, PasswordHash, Role, ContactInformation, AccountStatus, CreatedAt, UpdatedAt.

6.2.2. Event Data Detail

EventID, Title, Description, Category, Date, StartTime, EndTime, Venue, OrganizerID, Capacity, RegistrationStart, RegistrationEnd, Status, CreatedAt, UpdatedAt.

6.2.3. Registration Data Detail

RegistrationID, EventID, UserID, RegistrationDate, RegistrationStatus.

6.2.4. Attendance Data Detail

AttendanceID, EventID, UserID, AttendanceStatus, AttendanceDateTime.

The database should enforce primary-key uniqueness and appropriate foreign-key relationships. A

participant should not have duplicate active registration records for the same event.

7. Appendices

Appendix A – Proposed Technology Architecture: Browser-based client, application/server layer, and relational database layer. Exact programming language, framework, database product, and hosting provider are to be selected and documented by the project team.

Appendix B – Future Enhancements: QR-code attendance, email/SMS notifications, calendar integration, event feedback, automated certificates, and analytics.

8. Index

Attendance – 3.1.3, 6.1.3
Authentication – 3.1.1, 3.2.1, 5.2.1
Event Management – 3.1.2, 6.1.2
Registration – 3.1.3, 5.2.2, 6.1.3
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Reports – 3.1.4, 6.1.4

9. Annexes

9.1. Data flow diagram (optional)

The proposed data flow begins with the user authenticating through the system. Administrators and organizers submit or manage event information, which is stored in the event database. Participants retrieve published events and submit registration requests. Registration data is stored and later used by organizers to record attendance and generate reports.

9.2. Class diagram

Recommended classes: User, Role, Event, Registration, Attendance, Announcement, and Report. User has role-based relationships; Event is associated with an Organizer, Registration associates User and Event, and Attendance associates User and Event.

9.3. Use case realization (Sequence diagram / Communication diagram)

Primary realization: Participant → Event List → Event Details → Registration Service → Registration Database → Confirmation. Administrator realization: Administrator → Event Review → Approval Service → Event Database → Published Event.

9.4. User interface design

Recommended screens include Login, Dashboard, Event List, Event Details, Event Creation/Edit, Event Approval, Registration Management, Attendance Management, Announcements, User Management, and Reports.

9.5. Entity-relationship diagram

Core relationships: User 1-to-many Event as organizer; User 1-to-many Registration; Event 1-to-many Registration; User 1-to-many Attendance; Event 1-to-many Attendance; Event 1-to-many Announcement.